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To: The Editor-in-Chief and Editorial Board, *Biophysical Journal* (Cell Press / Biophysical Society)

Subject: Submission of Regular Article — *'Three-Dimensional Contact Topology Predicts Kirchhoff Contact Laplacian Fundamental Frequencies Across 1,040 Protein Structures'*

Dear Editor-in-Chief and Members of the Editorial Board,

Please find enclosed our original Regular Article, *'Three-Dimensional Contact Topology Predicts Kirchhoff Contact Laplacian Fundamental Frequencies Across 1,040 Protein Structures,'* by Alan Phipps, for consideration in the **Computational Biophysics / Protein Dynamics** section of *Biophysical Journal*. The review manuscript (biophysical_journal_manuscript.pdf, single-column with continuous line numbering), high-resolution white-background Figure 1 (fig1_n1040_headline_scaling_law.png / .pdf / .tif), standalone Supporting Material (Supporting_Information.pdf), and completed Cell Press Declaration of Interests Form are provided in full compliance with *Biophysical Journal* guidelines.

Scientific Motivation & Principal Findings:

While Gaussian Network Models (GNM) and Kirchhoff contact Laplacians have long served as foundational tools in macromolecular mechanics, a quantitative multi-scale benchmark testing whether a biomolecule's Kirchhoff contact Laplacian (dw-GNM) fundamental collective breathing mode (ν_1) is governed primarily by one-dimensional polypeptide chain length ($N_{C\alpha}$) or three-dimensional contact packing density ($\langle z \rangle$ / (R_g / R_0), with $R_0 \equiv 1.0$ Å) has lacked large-cohort cross-kingdom validation. Across N = 11 benchmark proteins, N = 115 primary validation structures, and N = 1,040 experimentally resolved PDB structures, oligomeric assemblies, and domain constructs (resolution ≤ 2.5 Å; non-redundant monomer chains culled at ≤ 25% sequence identity), we establish four principal findings:

1. Five-Fold Explanatory Advantage of 3D Contact Topology: Dimensionless 3D contact density $\langle z \rangle$ / (R_g / R_0) explains **80.48%** of the variance in the dw-GNM fundamental mode ν_1 ($r = 0.8953$, $R^2 = 0.8048$, $t = 64.74$, $p = 0.0002$), outperforming inverse radius $1/R_g$ (66.28%) and explaining five times more variance than 1D chain length $1/\sqrt{N_{C\alpha}}$ (16.02%).

2. Dimensionless Empirical Scaling Law & Cross-Kingdom Transfer: The relationship converges to $\nu_1(\text{THz}) = a_0(\langle z \rangle / (R_g / R_0))^b$ with $a_0 = 0.42338$ THz and $b = 1.1194$, maintaining accuracy under 10-fold cross-validation ($R^2_{CV} = 0.8039$), 80/20 holdout evaluation ($R^2_{OOS} = 0.8052$), and blind cross-kingdom transfer ($R^2_{OOS} = 0.8612$).

3. Continuum Elasticity Grounding of Functional Regimes: Globular catalytic enzymes (N = 922) exhibit super-linear scaling ($b = 1.1570$, $R^2 = 0.7992$) driven by interior rigidity percolation above the Maxwell threshold, whereas elongated structural/fibrillar proteins (N = 118) exhibit sub-linear filamentous scaling ($b = 0.9037$, $R^2 = 0.8701$).

4. Necessity of the Full Kirchhoff Laplacian (Γ_{ij}): We show that the remaining 19.52% residual variance reflects physical inter-domain hinge bottlenecks (3.179× frequency splitting between PDB ID 5O3L [letter O] and 1E9Z at 0.23% matched compactness), spectral anisotropy (λ_2 / λ_1, $r = -0.4521$), and crystallographic B_{iso} fluctuations ($r = 0.4257$).

Declaration of Interests: The author declares no competing financial, personal, or commercial interests related to this work. All computational analyses, eigensolvers, and model developments were performed independently by the author with zero external corporate sponsorship.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All data, models, and Python solvers are publicly archived under deterministic seed 20261018 with cryptographic SHA-256 manifests.

Sincerely,

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